

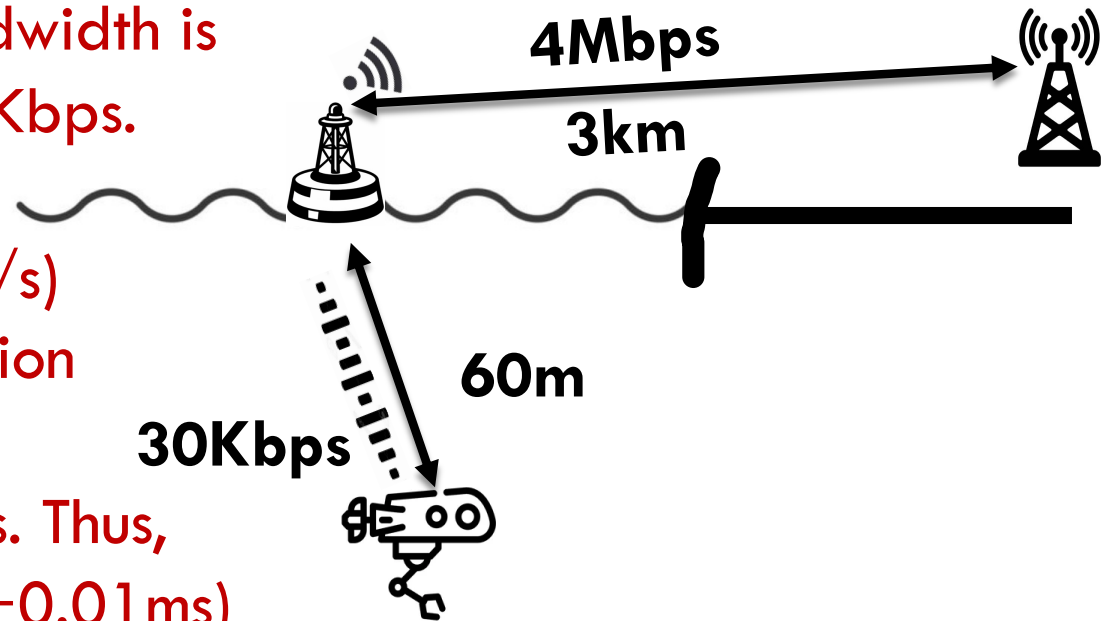
Quiz

46

- Let's define bandwidth-latency product (BDP) as the multiplication of the link bandwidth and the round-trip time delay. For the midterm network on the right, what is the BDP? Speed of sound is 1.5km/s , and speed of light is $3 \times 10^8\text{m/s}$.

Answer: the average bandwidth is $\min(30\text{Kbps}, 4\text{Mbps}) = 30\text{Kbps}$.

The propagation delay of first hop is $60\text{m}/(1.5\text{km/s}) = 40\text{ms}$, and the propagation delay of the 2nd hop is $3\text{km}/(3 \times 10^8\text{m/s}) = 0.01\text{ms}$. Thus, $\text{BDP} = 30\text{kbps} \times 2 \times (40\text{ms} + 0.01\text{ms}) \approx 2400 \text{ bits}$



Who Am I: Acquiring an IP Address

47

□ Dynamic Host Configuration Protocol (DHCP)

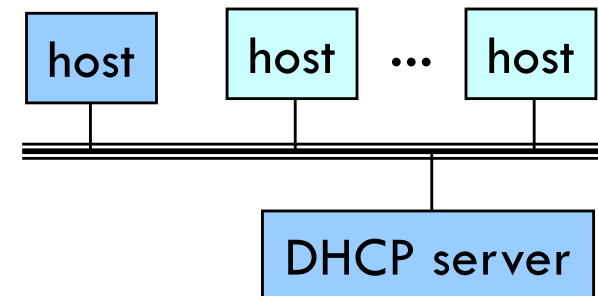
- Broadcast “I need an IP address, please!”
- Response “You can have IP address 1.2.3.4.”



DHCP

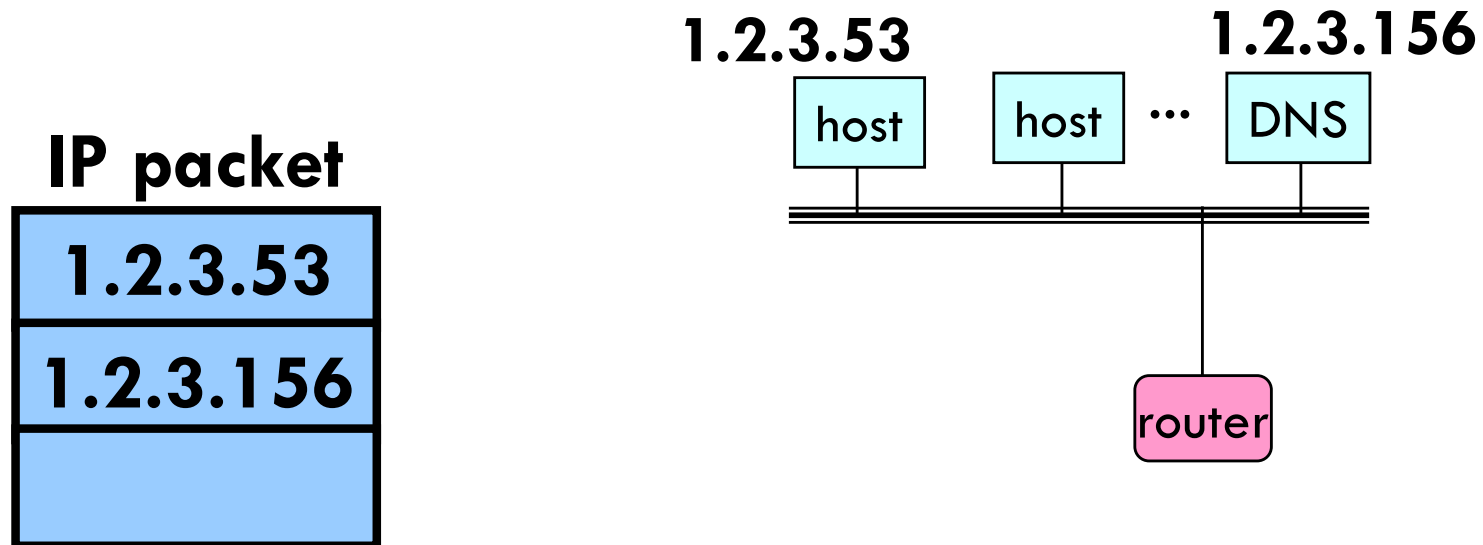
48

- Configure several parameters of hosts
 - ▣ Temporary IP address
 - ▣ Local DNS name server, gateway router, netmask
- DHCP server allocates addresses across users
- Broadcast a server-discovery message, server sends a reply offering an address



The Need of Address Resolution

50



- Adapters only understand MAC addresses
 - ▣ Translate the destination IP address to MAC address

Who Are You: Discovering the Receiver

51

□ Address Resolution Protocol (ARP)

- Broadcast “Who has IP address 1.2.3.6?”
- Response “0C-C4-11-6F-E3-98 has 1.2.3.6!”

DHCP server



71-65-F7-2B-08-53
1.2.3.4



1.2.3.6
0C-C4-11-6F-E3-98



1A-2F-BB-76-09-AD

ARP

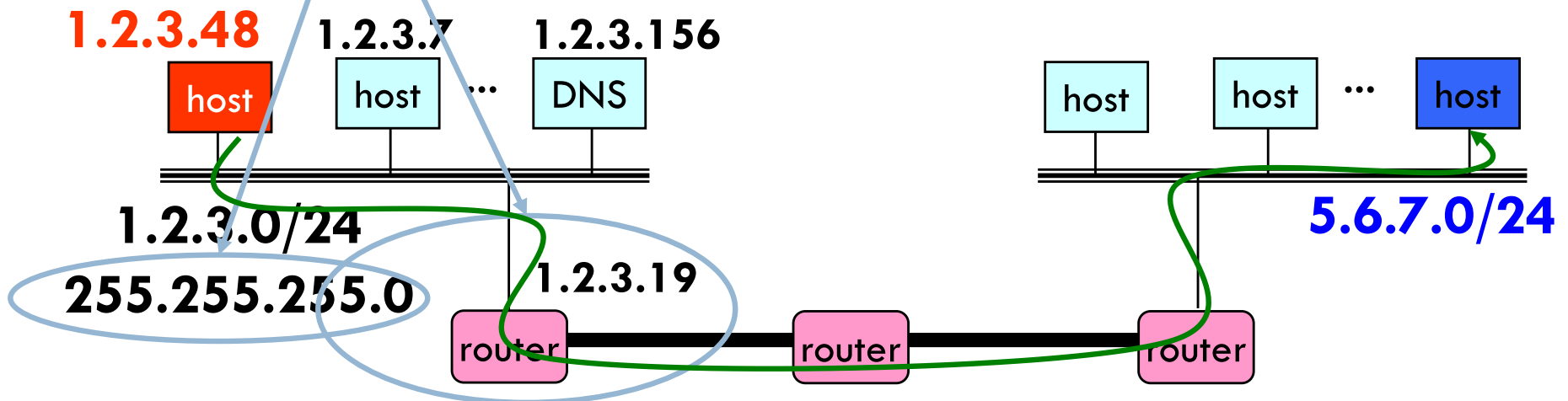
52

- Every node maintains an **ARP** table
 - ▣ <IP address, MAC address> pair, TTL
- Consult the table when sending a packet
 - ▣ Map destination IP address to destination MAC address
 - ▣ Encapsulate and transmit the data packet
- But: what if IP address **not** in the table?
 - ▣ Sender **broadcasts**: “**Who has IP address 1.2.3.156?**”
 - ▣ Receiver responds: “**MAC address 58-23-D7-FA-20-B0**”
 - ▣ Sender **caches** result in its ARP table

What if Remote Destination?

53

- Look up the MAC address of the first hop router
 - ▣ 1.2.3.48 uses ARP to find MAC address for first-hop router **1.2.3.19** rather than ultimate destination IP address
- How does the host know the destination is not local?
 - ▣ Uses netmask (discovered via DHCP)
- How does the red host know about 1.2.3.19?
 - ▣ Also DHCP



Key Ideas in ARP and DHCP

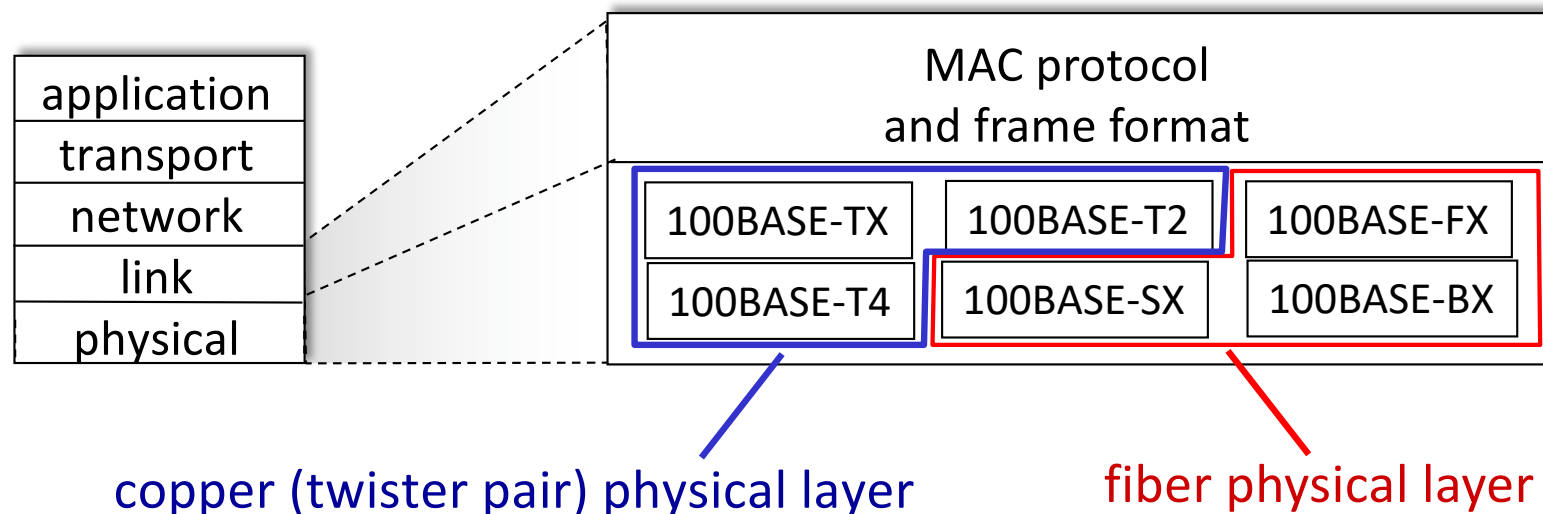
54

- **Broadcasting:** Can use broadcast to make contact
 - ▣ Scalable because of limited size
- **Caching:** remember the past for a while
 - ▣ Store the information you learn to reduce overhead
 - ▣ Remember your own address & other host's addresses
- **Soft state:** eventually forget the past
 - ▣ Associate a **time-to-live** field with the information
 - ▣ ... and either refresh or discard the information
 - ▣ Key for **robustness** in the face of unpredictable change

802.3 Ethernet standards: link & physical layers

55

- Many different Ethernet standards
 - ▣ Common MAC protocol and frame format
 - ▣ Different speeds: 2 Mbps, ... 100 Mbps, 1 Gbps, 10 Gbps, 40 Gbps, 80 Gbps
 - ▣ Different physical layer media: fiber, cable



Switches

56

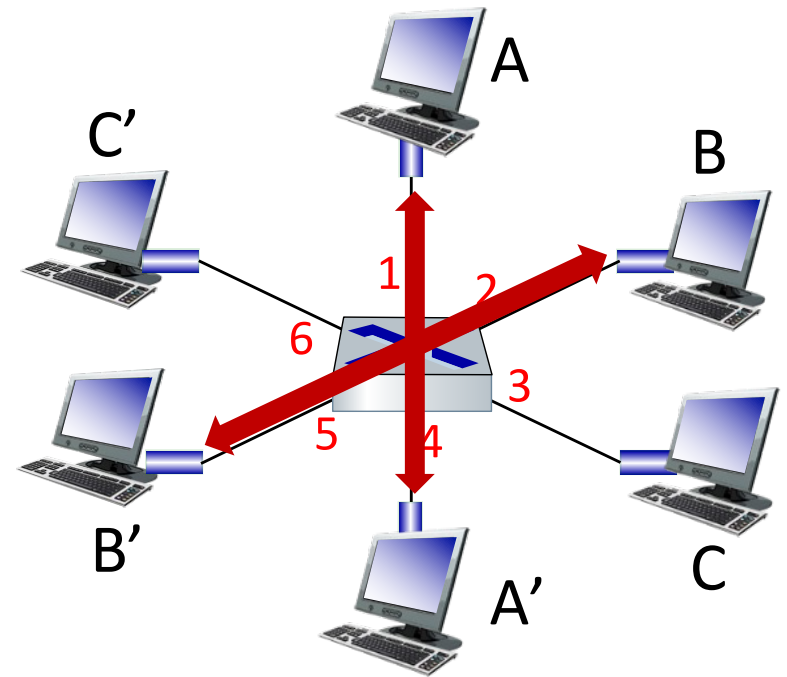
- Switch is a link-layer device
 - store, forward Ethernet (or other type of) frames
 - examine incoming frame's *MAC* address, selectively forward frame to one-or-more outgoing links device
- plug-and-play, self-learning
 - switches do not need to be configured

Switch:

Multiple simultaneous transmissions

57

- hosts have dedicated, direct connection to switch
- switches buffer packets
- Ethernet protocol used on each incoming link, so:
 - ▣ no collisions; full duplex
 - ▣ each link is its own collision domain



switch with six interfaces
(1,2,3,4,5,6)

*A-to-A' and C-to-A' can't happen
simultaneously*

Switch Forwarding Table

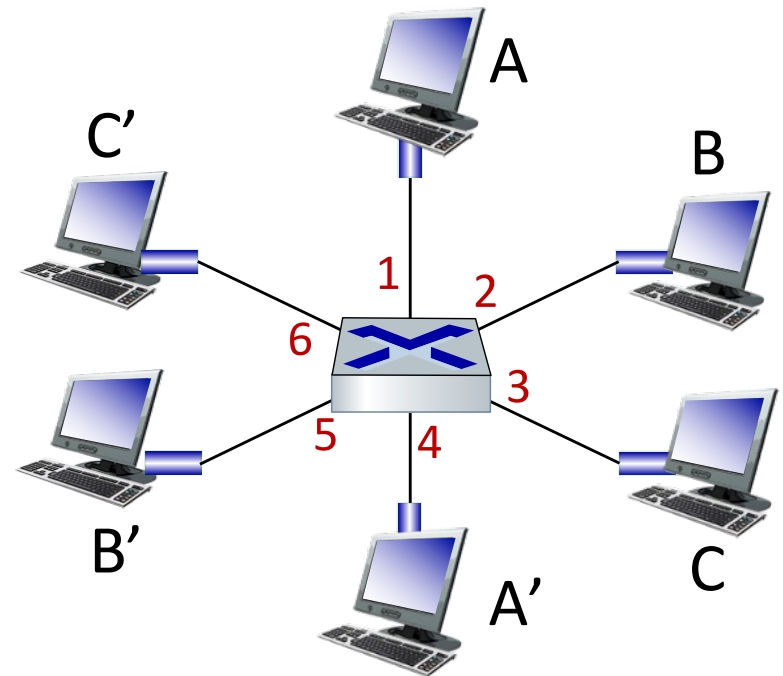
58

Q: how does switch know A' reachable via interface 4, B' reachable via interface 5?

A: each switch has a **switch table**, each entry:

- (MAC address of host, interface to reach host, time stamp)
- looks like a routing table!

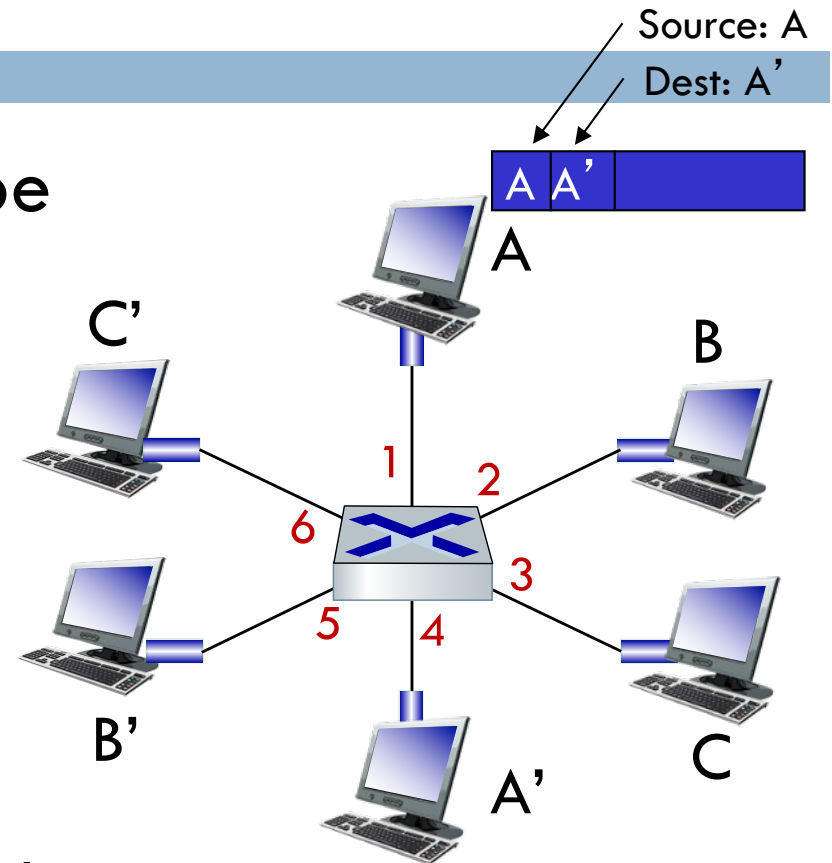
Switch table is **self-learned**



Switch: Self Learning

59

- switch learns which hosts can be reached through which interfaces
- ▣ when frame received, switch “learns” location of sender
- ▣ records sender/location pair in switch table
- ▣ Unknow frame destination: flood
- ▣ Know destination: send on just one link



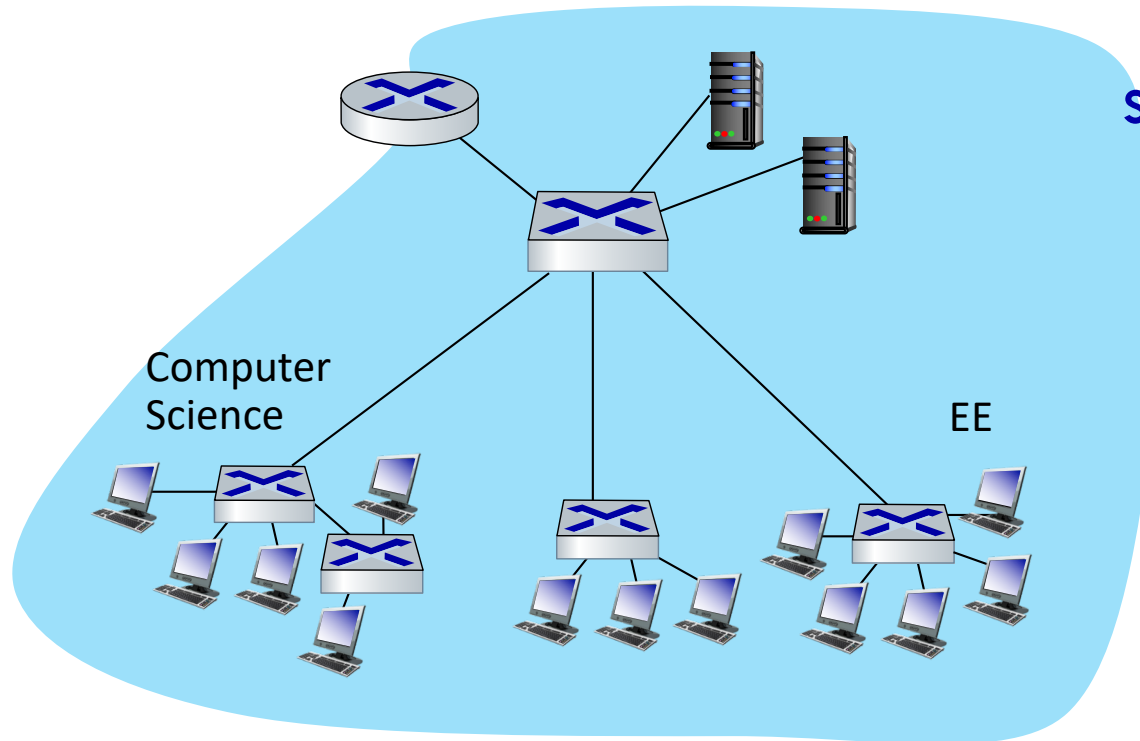
MAC addr	interface	TTL
A	1	60

Switch table
(initially
empty)

Virtual LAN (VLAN)

60

- **Q:** what happens as LAN sizes scale, users change point of attachment?



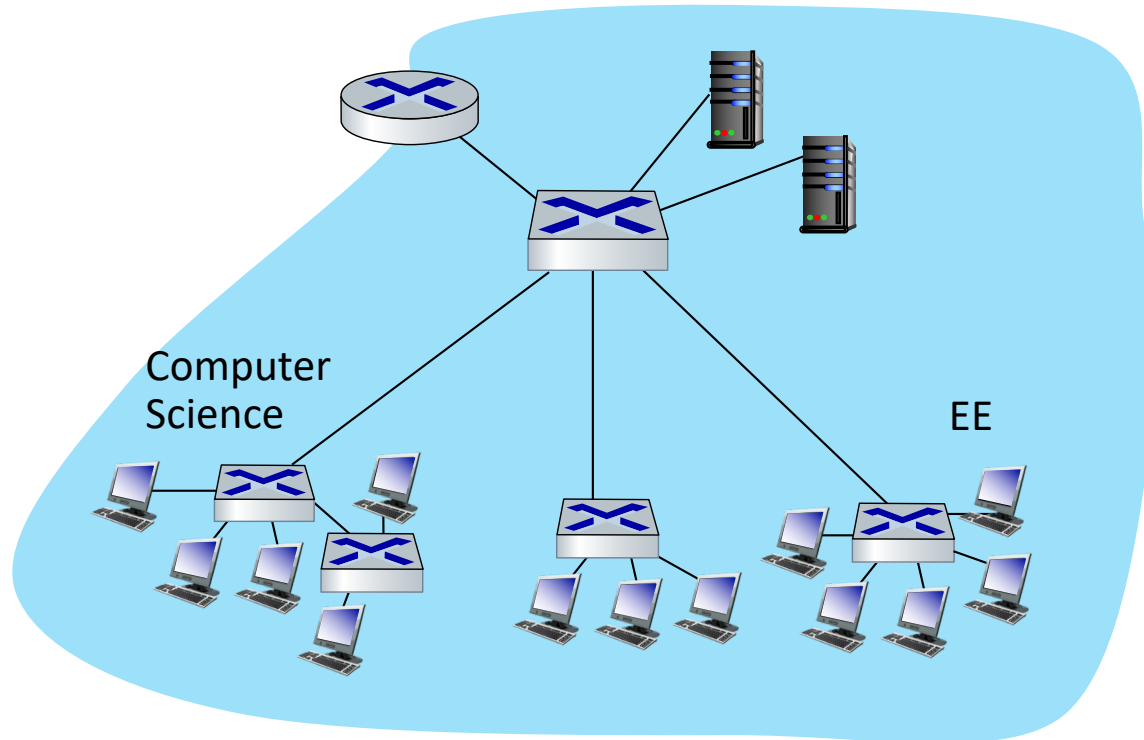
single broadcast domain:

- *scaling*: all layer-2 broadcast traffic (ARP, DHCP, unknown MAC) must cross entire LAN
- efficiency, security, privacy issues

Virtual LAN (VLAN)

61

- **Q:** what happens as LAN sizes scale, users change point of attachment?



administrative issues:

- CS user moves office to EE - *physically* attached to EE switch, but wants to remain *logically* attached to CS switch

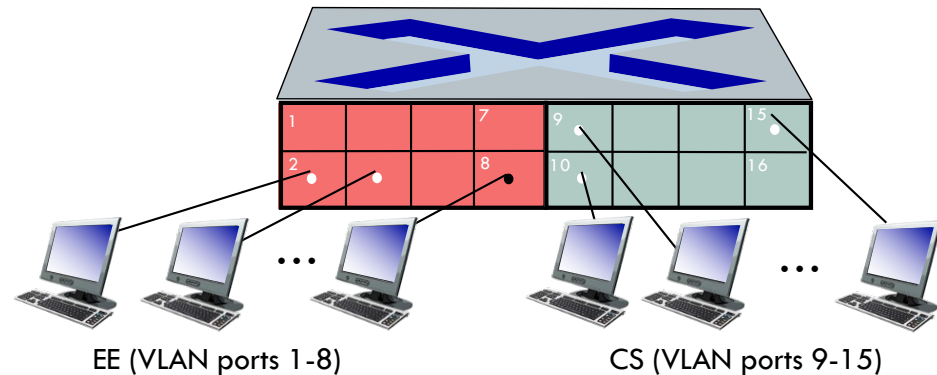
Port-based VLAN

62

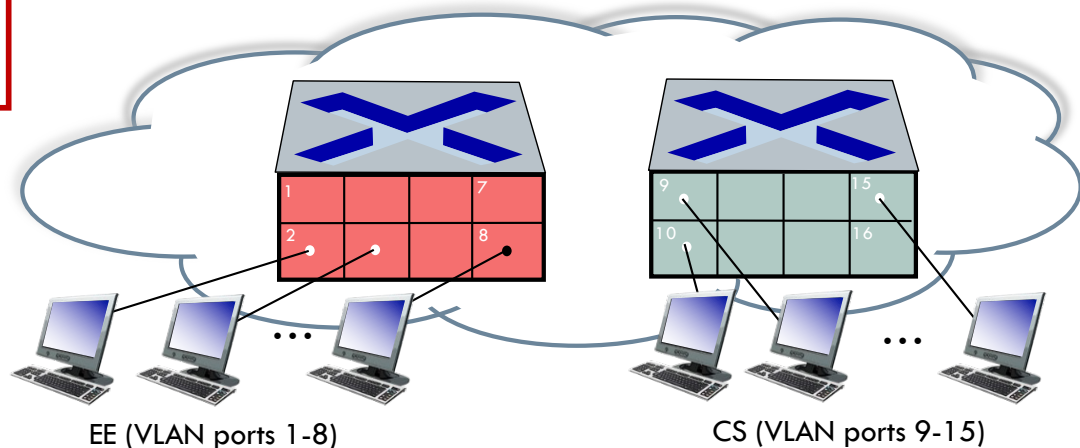
port-based VLAN: switch ports grouped (by switch management software) so that *single* physical switch
.....

Virtual Local Area Network (VLAN)

switch(es) supporting VLAN capabilities can be configured to define multiple *virtual* LANS over single physical LAN infrastructure.



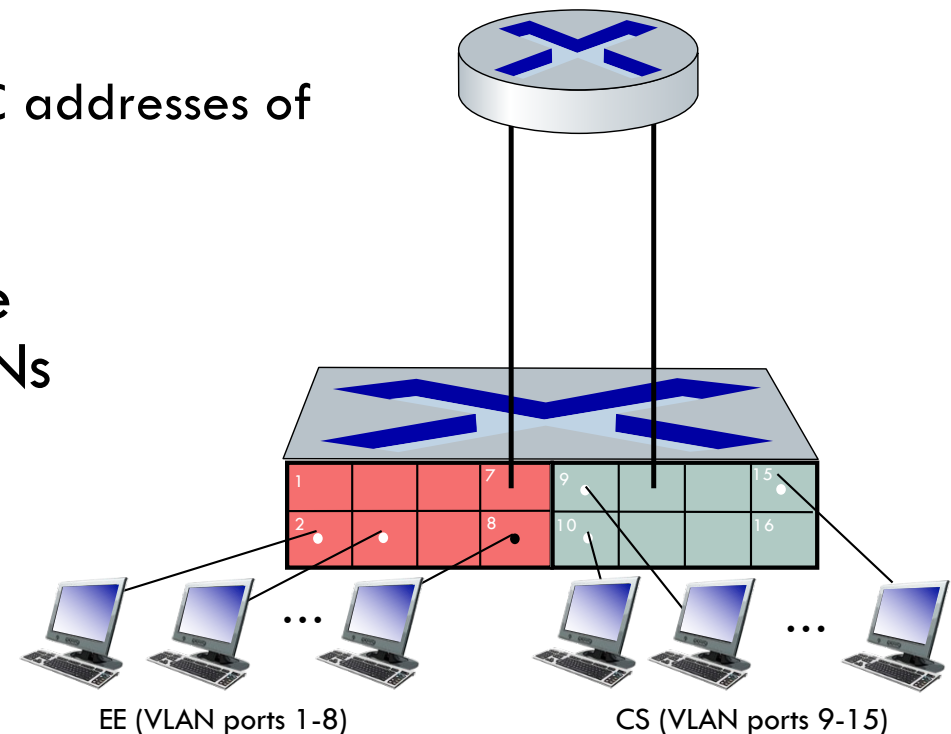
... operates as **multiple** virtual switches



Port-based VLAN

63

- **traffic isolation:** frames to/from ports 1-8 can *only* reach ports 1-8
 - can also define VLAN based on MAC addresses of endpoints, rather than switch port
- **dynamic membership:** ports can be dynamically assigned among VLANs
- **forwarding between VLANs:** done via routing (just as with separate switches)
 - in practice vendors sell combined switches plus routers



Routers vs Switches

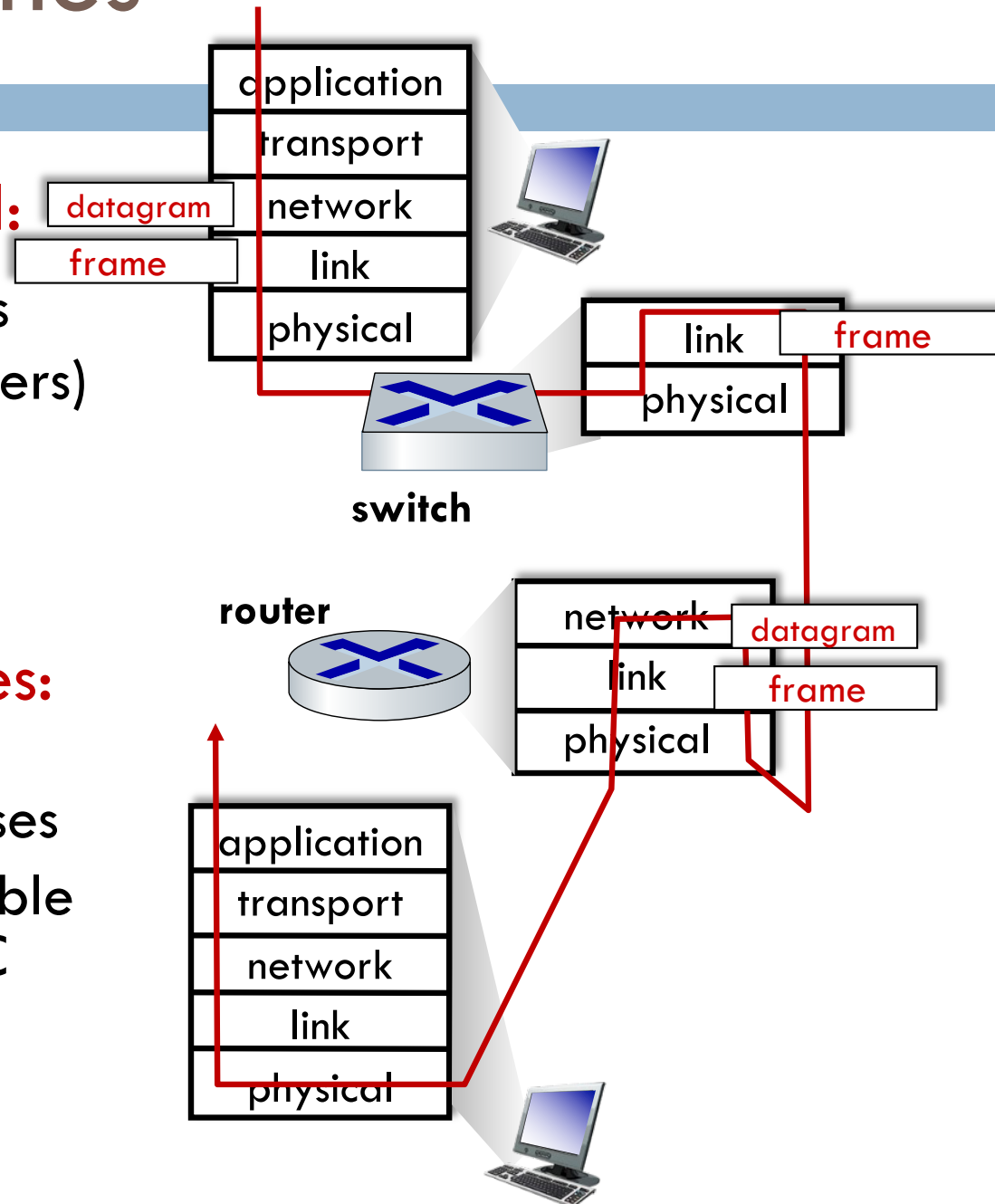
64

both are store-and-forward:

- routers: network-layer devices (examine network-layer headers)
- switches: link-layer devices (examine link-layer headers)

both have forwarding tables:

- routers: compute tables using routing algorithms, IP addresses
- switches: learn forwarding table using flooding, learning, MAC addresses



CSEE4199: COMPUTER NETWORK



WIRELESS PHYSICAL LAYER & MAC PROTOCOLS

April 9, 2026

Xia Zhou

My Goals

2

- Make sure you know basic physics of wireless communications and basic knowledge on wireless MAC protocols



- Expose you to super interesting research on wireless networking

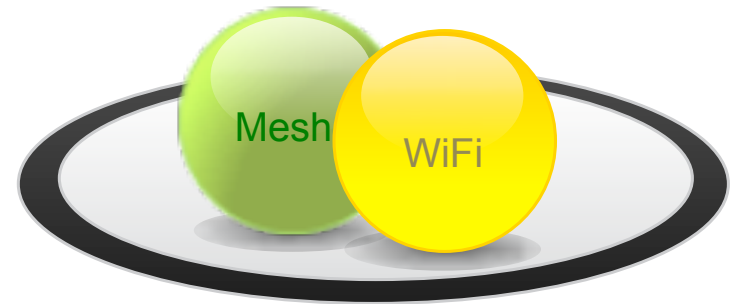


Wireless Networks

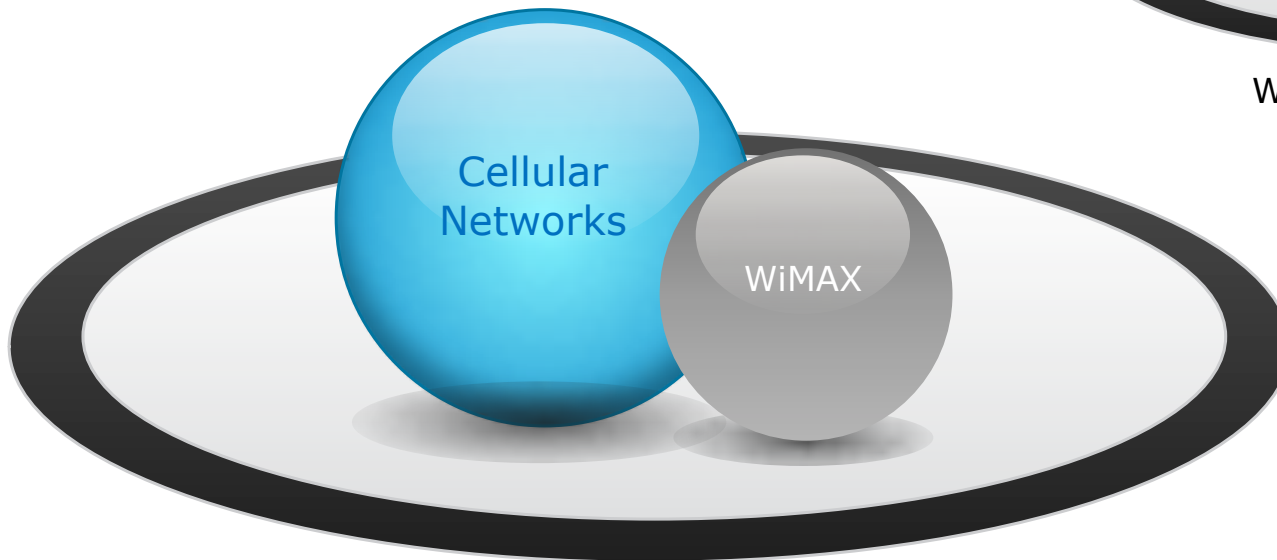
3



WPAN: Personal Area Networks

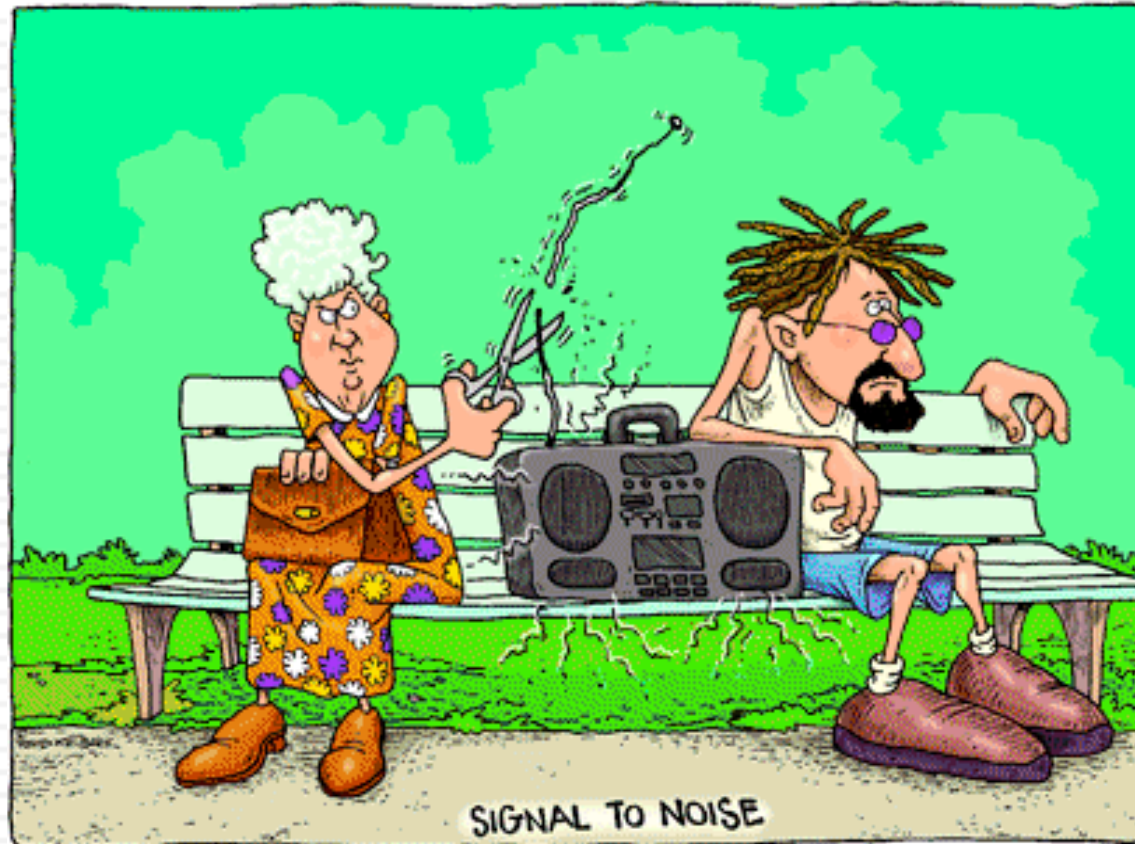


WLAN: Local Area Networks



WWAN: Wide Area Networks
Centralized Networks for Tight Control and
Seamless Connectivity

Wireless Signal Propagation 101



If We can See the Wi-Fi Signals...

5



<http://gizmodo.com/what-the-world-would-look-like-if-you-could-actually-se-846030448>

If We can See the Wi-Fi Signals...

6



<http://gizmodo.com/what-the-world-would-look-like-if-you-could-actually-se-846030448>

If We can See the Wi-Fi Signals...

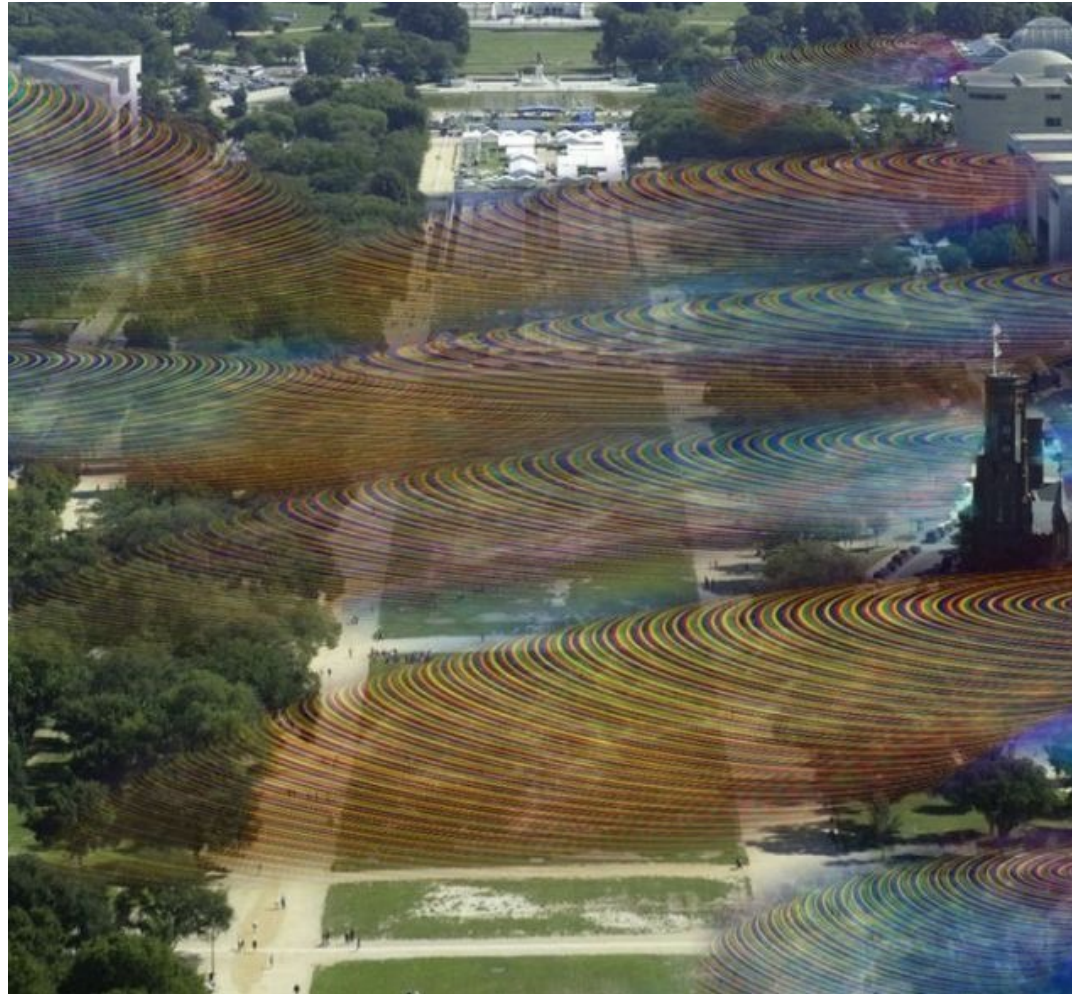
7



<http://gizmodo.com/what-the-world-would-look-like-if-you-could-actually-se-846030448>

If We can See the Wi-Fi Signals...

8



<http://gizmodo.com/what-the-world-would-look-like-if-you-could-actually-se-846030448>

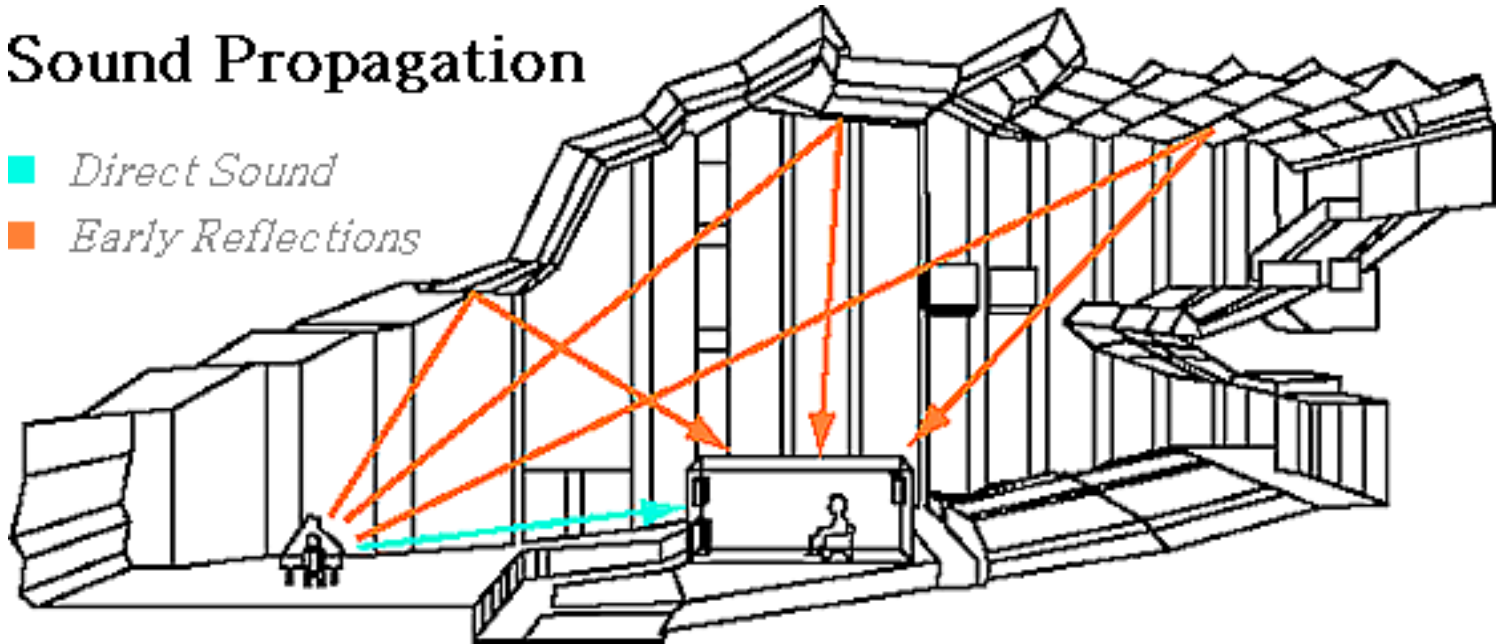
What Happens if We Cut the Wire

9



Sound Propagation

- *Direct Sound*
- *Early Reflections*



Just Consider the Direct Path

10

□ Free-space pathloss

- Transmission power decreases with **square** of distance

$$\text{Signal power loss} = \left(\frac{4\pi d}{\lambda} \right)^2 = \left(\frac{4\pi d f}{c} \right)^2$$

- d: distance between TX and RX
- λ : signal wavelength in meters
- f: signal frequency in hertz
- c: speed of light in a vacuum

Higher frequency (shorter wavelength) has
higher signal loss

Just Consider the Direct Path

11

□ Free-space pathloss

- Transmission power decreases with **square** of distance

$$\text{Signal power loss} = \left(\frac{4\pi d}{\lambda} \right)^2 = \left(\frac{4\pi d f}{c} \right)^2$$

Freq.	900MHz	2.4GHz	5GHz
Range	902 – 928	2.4 – 2.4835	5.15 – 5.35
Bandwidth	26MHz	83.5MHz	200MHz
Wavelength	0.33m / 13.1"	0.125m / 4.9"	0.06m / 2.4"

Just Consider the Direct Path

12

- In general cases

$$\text{Signal power loss} = \left(\frac{4\pi df}{c} \right)^\alpha$$

← Pathloss component

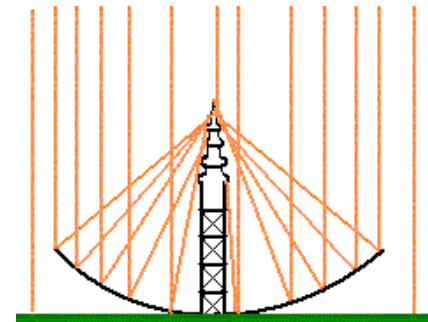
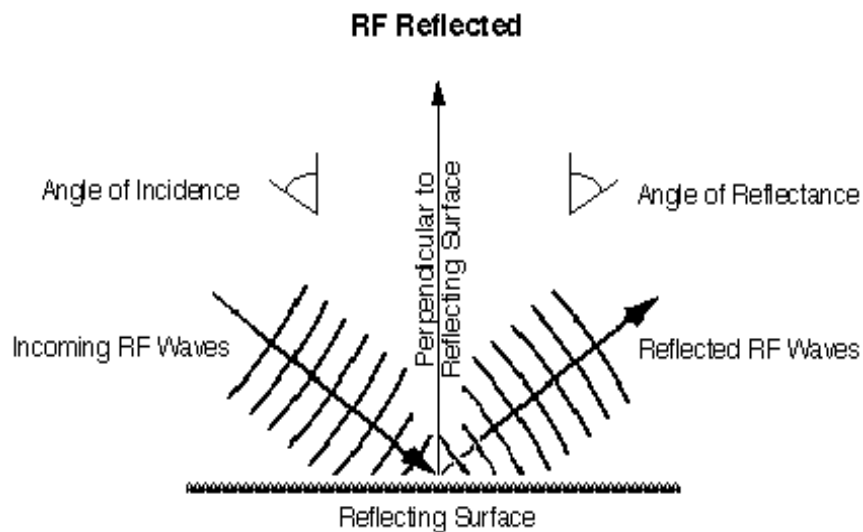
Free space	2
Urban area cellular	2.7 – 3.5
Shadowed urban cell	3 – 5
Obstructed in building	4 – 6
Obstructed in factories	2 – 3

Signal Propagation

13

□ Reflection

- Propagation wave impinges on an object larger than its wavelength, e.g., buildings, walls



**Boost WiFi signals
using a beer can!**

